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Integrated Multi-Criteria Analysis of Water Stress in Al Hoceima Province, Morocco

Ouassima El Bannoudi [1], Bojan Đurin [2,*], Hala Rossi [3], Hakim Boulaassal [3] and Khadija Haboubi [1]

[1] Laboratory of Engineering Sciences and Applications, National School of Applied Sciences of Al Hoceima, Abdelmalek Essâadi University, Al-Hoceima 32003, Morocco; ouassima.elbannoudi@etu.uae.ac.ma (O.E.B.); k.haboubi@uae.ac.ma (K.H.)
[2] Department of Civil Engineering, University North, 48000 Koprivnica, Croatia
[3] Research and Development in Applied Geosciences Laboratory, Faculty of Sciences and Techniques of Tangier, Abdelmalek Essaadi University, Tetouan 90000, Morocco; hala.rossi@etu.uae.ac.ma (H.R.); h.boulaassal@uae.ac.ma (H.B.)
* Correspondence: bdjurin@unin.hr

Abstract

Water scarcity is becoming an increasingly serious environmental and socio-economic problem, particularly in semi-arid regions where water resources are increasingly stressed by climate variability and growing human pressure. Morocco is one of the most water-stressed countries in the Mediterranean region due to decreased rainfall, higher temperatures, and population growth. The aim of this study is to evaluate the dynamics of water stress in Al Hoceima Province using an integrated framework that combines hydrological indicators and Multi-Criteria Decision Making (MCDM) methods. This locally focused assessment integrates hydro-climatic and demographic indicators to provide a better understanding of water stress dynamics in the province. Precipitation, temperature, runoff, evaporation, and population are considered. The Water Stress Index (WSI) and Falkenmark Index were used to determine water stress conditions. The weights of each criterion were assigned using the Analytic Hierarchy Process (AHP), while annual water stress conditions were analyzed using the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) and the Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE). Additionally, Innovative Trend Analysis (ITA) was used to analyze long-term trends in water stress. Results indicate that water stress was consistently high throughout the study period and highly variable between years, primarily due to hydro-climatic conditions. Runoff and precipitation were the most important factors, while higher temperatures and evaporation exacerbated water stress. TOPSIS and PROMETHEE showed consistent identification of critical wet and dry years.

Keywords: water scarcity; Water Stress Index (WSI); Falkenmark Index; multi-criteria decision-making; hydro-climatic variability

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1. Introduction

Water scarcity has become one of the most significant environmental and socio-economic challenges of the 21st century [1]. Increasing freshwater demand, driven by population growth and climate change, is exerting growing pressure on global freshwater resources [1]. Consequently, a significant percentage of the global population is already exposed to water stress, and this is bound to increase over the next few decades. Sustainable water resource management thus necessitates integrated water management,

which incorporates improved monitoring, more efficient water use, and the development of decision-support tools that can capture the complexity of environmental systems [1]. On the international level, socio-economic development and climate change interact to produce water stress. The growing population and economic activity are set to increase water demand worldwide, especially in developing nations. It is estimated that by 2050, socio-economic factors alone may result in an increasing number of people living in conditions of at least moderate water stress [2]. This situation is further exacerbated by climate change, which alters major hydrological processes, including precipitation distribution, runoff, and evaporation rates. Despite its geographically variable effects, climate change is likely to increase water-stress vulnerability in most areas of the planet and introduce high uncertainty about future water supply [3]. Especially sensitive to the changes are countries in arid and semi-arid regions. Water resources are significantly affected by climatic variability and frequent droughts in the Mediterranean Basin and North Africa. Morocco is experiencing increased pressure on water systems, particularly due to declining rainfall, rising temperatures, and increasing water demand. Recent reports also highlight persistent drought conditions and water-resource pressures in the Al Hoceima region, particularly during the 2023–2024 hydrological year [4]. According to hydrometeorological measurements, 2022 was the hottest year in the country's history, with temperatures above the annual average for about 80% of the year and several national monthly temperature records broken [5]. Meanwhile, precipitation was about 27% lower than average, and 2022 was the twelfth-driest year in Morocco. Moreover, the four-year period 2019–2022 will be the driest since the 1960s [5]. These trends demonstrate the further worsening of the situation with droughts and emphasize the further susceptibility of Moroccan water resources. As a result, Morocco is now thought to be among the most water-stressed nations in the Middle East and North Africa region. Water scarcity in the country is a cumulative effect of climate change, over-use of water resources, and uneven spatial distribution of water resources in various regions [6]. These stressors endanger water security and pose great challenges to sustainable development, agriculture, and ecosystem protection. Due to the dynamic nature of water scarcity, water management requires analytical methods that can incorporate various environmental and socio-economic factors. Decision-support tools are thus important for assisting policymakers and water managers by supplying quantitative data to help them assess various management approaches and make informed future decisions [7]. Among these strategies, multi-criteria decision-making (MCDM) methods have received increasing attention over the last few years, as they allow the incorporation of various indicators and variables into a single analytical framework. Some techniques, such as the Analytic Hierarchy Process (AHP), the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), and the Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE), are widely used to assign weights to criteria and rank alternative scenarios within complex environmental systems [8–10].

Despite the growing literature on water scarcity and water resource management, a gap remains in integrating hydrological indicators with multi-criteria decision-making models to assess water stress, not only at the regional scale in Morocco but also worldwide. In particular, the reviewed literature did not identify a comprehensive study applying such an integrated framework in Al Hoceima Province, despite increasing climatic variability and growing pressure on regional water resources. This gap limits the understanding of local water stress dynamics and highlights the need for more effective management strategies.

Accordingly, this study aims to assess water stress in Al Hoceima Province using an integrated framework that combines hydrological indicators with multi-criteria decision-making (MCDM) approaches. More specifically, the analysis integrates the Falkenmark Index and the Water Stress Index (WSI) with the Analytic Hierarchy Process (AHP) for

criteria weighting, while the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) and the Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE) are employed to rank annual water stress conditions. In addition, the Innovative Trend Analysis (ITA) method is applied to examine long-term trends in water stress, providing a comprehensive assessment of the temporal dynamics of water resources in the study area [11].

This study's methodological contribution does not rely on developing new individual MCDM methods, but on their integrated application within a common water-stress assessment framework. Research presented in [12] used SAW, TOPSIS, and PROMETHEE-II methods for groundwater resource management. Multicriteria methods are widely used in ranking and decision-making processes in water management [13]. The ITA method examines trends in time-series data. Examples in [14,15] show applications that provide efficient insight into the properties of hydrological data. In the present study, we combine these established methods to provide a decision-support framework for Al Hoceima Province.

The study also contributes a 45-year dataset (1981–2025), allowing water stress to be examined at annual, monthly, seasonal, and long-term temporal scales. The combination of the Falkenmark Index, WSI, AHP-based weighting, TOPSIS and PROMETHEE ranking, ITA trend analysis, and sensitivity analysis provides a complementary decision-support interpretation of water-stress dynamics in the study area.

2. Materials and Methods

2.1. Falkenmark Water Stress Indicator

Water availability was assessed using the Falkenmark water stress indicator, which estimates the amount of renewable freshwater per person [16]. In this study, annual runoff was used as a proxy for renewable water resources. The provincial area was used to convert runoff depth (mm) into volumetric water resources (m^3). The area of Al Hoceima Province is 3550 km^2, according to the official information provided by the High Commission for Planning (HCP) [17], the study area was extracted from the geographic dataset using a Python script developed and executed in PyCharm 2026.1.2 (JetBrains s.r.o., Prague, Czech Republic) with Python 3.14 (64-bit; Python Software Foundation). Annual runoff depth was converted into water volume by multiplying the runoff depth in metres by the provincial area in m^2. Water availability per capita was then calculated by dividing the annual renewable water resources by the corresponding population. The Falkenmark indicator could be formulated as:

$$\mathrm{FWI} = \frac{R \times A}{P} \quad (1)$$

R is the yearly runoff depth converted from mm to meters, A is the area of the province in m^2, and P is the annual population.

This is a widely used indicator for assessing water stress and offers an easy method for assessing water availability relative to population pressure [16]. The classification limits related to the Falkenmark indicator are outlined in Table 1, which depicts the various types of water availability based on per capita renewable water resources [18].

Table 1. Falkenmark (1989) classification of water stress levels.

Category	m^3/Year/Capita
No stress	>1700
Stress	1000–1700
Scarcity	500–1000
Absolute scarcity	<500

2.2. Criteria Weighting Using the Analytic Hierarchy Process (AHP)

The Analytic Hierarchy Process (AHP) proposed by Thomas L. Saaty [8] was used to determine the relative importance of the identified variables. AHP is a popular multi-criteria decision-making method that allows the assessment of quantitative and qualitative criteria through a hierarchical structure, pairwise comparisons, and priority synthesis [19].

In this approach, a pairwise comparison matrix was developed to determine the relative importance of runoff, precipitation, evaporation, temperature, and population in relation to water stress.

This study considered five criteria: runoff, precipitation, evaporation, temperature, and population. The pairwise comparison matrix was established using Saaty's fundamental 1–9 scale.

The judgments reflected each criterion's relative importance in representing water stress, based on the authors' hydrological knowledge and the observed statistical relationships between the criteria and the Falkenmark Index. The correlation analysis supported the relative importance of the criteria and was not mechanically converted into Saaty scores.

The pairwise comparison matrix A = $[a_{ij}]$ is given by:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \\ a_{41} & a_{42} & a_{43} & a_{44} & a_{45} \\ a_{51} & a_{52} & a_{53} & a_{54} & a_{55} \end{bmatrix}$$

a_{ij} is the importance of criterion i over criterion j. The diagonal terms are unity ($a_{ii} = 1$), and the matrix satisfies the reciprocal property $a_{ji} = 1/a_{ij}$.

The authors assigned the pairwise comparison values based on their hydrological knowledge and their interpretation of the statistical relationships between the selected criteria and the Falkenmark Index, without involving external experts in a formal expert elicitation process. Table 2 presents the complete numerical pairwise comparison matrix used in this study.

Table 2. AHP pairwise comparison matrix for the five water stress criteria.

AHP	Runoff	Precipitation	Evaporation	Temperature	Population
Runoff	1	3	5	5	7
Precipitation	1/3	1	3	3	5
Evaporation	1/5	1/3	1	3	3
Temperature	1/5	1/3	1/3	1	3
population	1/7	1/5	1/3	1/3	1

Runoff was given the highest relative importance because it directly represents the amount of water generated and potentially available within the study area. It was therefore assigned a value of 5 compared with evaporation and temperature and 7 compared with population. Precipitation was also assigned a high importance because it is the main climatic input contributing to water availability. It was considered moderately more important than evaporation and temperature (value of 3 in both comparisons) and moderately more important than population (also value of 3). Evaporation was considered more important than temperature and population (value of 3) because it directly represents water loss from the system. Temperature was assigned a value of 3 relative to population because it is an important hydro-climatic stress factor, particularly through its influence on evaporative losses. Population received the lowest relative importance because it repre-

sents demographic pressure rather than direct water use, and the study included no direct water-consumption data. These comparisons were used to construct the AHP matrix and derive the final criterion weights. Table 3 summarizes each criterion's role and the rationale for its relative importance in the AHP weighting process.

Table 3. Role and relative importance of the criteria used in the AHP analysis.

Criterion	Hydrological Meaning	Relative Importance in AHP
Runoff	Direct indicator of water availability	Highest
Precipitation	Main climatic input contributing to water availability	High
Evaporation	Direct representation of water loss	Intermediate
Temperature	Hydro-climatic stress factor influencing evaporative losses	Lower
Population	Indirectly reflects pressure on available water resources	Lowest

The correlations with the Falkenmark Index were 0.999 for runoff, 0.850 for precipitation, 0.703 for evaporation, −0.251 for temperature, and 0.151 for population. These values are only used as supporting information and are not mathematically transformed into the 1–9 Saaty scale.

The matrix was then normalized by dividing each element by the sum of its column:

$$a_{ij}^{norm} = \frac{a_{ij}}{\sum_{i=1}^{n} a_{ij}} \quad (2)$$

The weight of each criterion was then calculated by taking the average of each row:

$$\omega_i = \frac{1}{n}\sum_{j=1}^{n} n_{ij} \quad (3)$$

ω_i is the weight of criterion i, and n is the number of criteria.

The maximum eigenvalue (λ_{max}) was calculated to check the validity of the pairwise comparisons. The consistency of judgments was measured with the Consistency Index (CI) and the Consistency Ratio (CR), which are calculated as follows:

$$\mathrm{CI} = \frac{\lambda_{max} - n}{n - 1} \quad (4)$$

$$\mathrm{CR} = \frac{CI}{RI} \quad (5)$$

λ_{max} is the maximum eigenvalue of the comparison matrix, n is the number of criteria, and RI is the Random Index depending on the size of the matrices.

RI is the Random Index based on the size of the matrices [8]. The matrix is considered consistent if CR < 0.1 (Saaty criterion) and the judgments are good.

The resulting weights were 0.4881 for runoff, 0.2400 for precipitation, 0.1359 for evaporation, 0.0897 for temperature, and 0.0463 for population. The maximum eigenvalue was 5.2845, with a CI of 0.0711 and a CR of 0.0635. A CR less than 0.1 indicates that the relative importance weights are consistent and reliable [8].

The weights were later used to develop the Water Stress Index (WSI) and to calculate TOPSIS to assess water stress.

Sensitivity Analysis of AHP Weights

A one-at-a-time sensitivity analysis [20] was conducted to assess the sensitivity of the AHP-derived weights. Each criterion weight was independently increased or decreased

by 10%, 20%, and 30%, while the other weights were initially kept at their original values. The resulting weight vector was then normalized so that the sum of all five weights equaled 1. The WSI was then recalculated for each perturbed weight set, resulting in 30 sensitivity scenarios.

2.3. Construction of the Water Stress Index (WSI)

The Water Stress Index (WSI) was developed to assess at both annual and monthly temporal scales by combining crucial hydro-climatic and socio-demographic factors, i.e., runoff (mm), precipitation (mm), evaporation (mm), temperature (°C), and population. All variables were normalized before aggregation to make them comparable across different units and scales through min–max normalization. Two normalization methods were used based on each variable's impact on water stress. The variables that led to higher water stress (evaporation, temperature, and population) were normalized as:

$$X_i = \frac{X_i - X_{min}}{X_{max} - X_{min}} \quad (6)$$

where variables associated with greater water availability (runoff and precipitation) were negatively normalized as:

$$X_i = \frac{X_{max} - X_i}{X_{max} - X_{min}} \quad (7)$$

This normalization ensures that higher normalized values are consistently associated with greater water stress [21]. The WSI was then calculated by linearly aggregating the normalized variables with weight:

$$WSI = \sum_{i=1}^{n} \omega_i . X_i \quad (8)$$

where ω_i is the weight of each variable, which is calculated using the Analytic Hierarchy Process (AHP), and Xi. This multi-criteria aggregation method is popular in water resource evaluation research because it enables the incorporation of heterogeneous indicators into a single composite index that represents the overall degree of water stress [21].

We calculated the monthly WSI using the same min–max normalization and weighted aggregation procedure as the annual WSI. The monthly dataset comprised 540 observations corresponding to 12 months for each year from 1981 to 2025. For each criterion, we determined the minimum and maximum values from the complete monthly dataset over the study period. We normalized runoff and precipitation using the inverse min–max formulation because higher values indicate greater water availability, whereas we normalized evaporation, temperature, and population using the direct min–max formulation because higher values contribute to greater water stress. The normalized monthly criteria were then aggregated using the same AHP-derived weights, producing one WSI value for each month. For the seasonal analysis, we averaged WSI values for the same calendar month over the 45-year period.

Validation of the WSI

To further assess WSI consistency, the annual WSI values were compared with the Falkenmark Index for 1981–2025. Since the two indices have opposite interpretations, Spearman's rank correlation coefficient (ρ) was used to examine their relationship. We used a significance level of 0.05.

2.4. Multi-Criteria Decision Analysis Using the TOPSIS Method

The Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) was used to classify the water stress conditions in Al Hoceima Province [9,22]. This is a multi-criteria

decision-making method that ranks the alternatives based on their relative nearness to an ideal scenario and their distance from the worst possible scenario. The alternatives in this study are the various years of the study period. The analysis accounted for five variables: runoff, precipitation, evaporation, temperature, and population. They were chosen based on hydrological logic and justified by the general principles of the hydrological cycle; precipitation, evapotranspiration, and runoff are the main components of water availability and loss [23]. To avoid repetition in the evaluation, the Falkenmark index was not included as an additional criterion, as it already accounts for runoff and population.

The annual values of the chosen variables [24] were used to create the decision matrix prior to applying the TOPSIS procedure. The variables have different units and scales; hence, the matrix was initially normalized to compare them [24]. The normalized decision matrix was obtained as follows:

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}} \quad (9)$$

x_{ij} is the value of criterion $_j$ for alternative i, m is the number of alternatives (years), and r_{ij} is the normalized value.

Each criterion guides the direction of influence, which the TOPSIS procedure clearly accounts for. The runoff and precipitation were considered benefit criteria, with higher values reflecting more favorable conditions for water availability, while the evaporation, temperature, and population were considered cost criteria, with higher values reflecting less favorable conditions for water availability. Hence, there was no reverse normalization in this stage. The classification of the five criteria as benefit or cost criteria, together with their interpretation, is summarized in Table 4.

Table 4. Classification of the criteria according to their influence on water stress.

Criterion	Type	Interpretation of Higher Value
Runoff	Benefit	Good water availability
Precipitation	Benefit	Good water availability
Evaporation	Cost	Less favorable water availability
Temperature	Cost	Less favorable water availability
Population	Cost	Growth increased demands on water resources

After normalization and weighting, the positive ideal solution was determined from the maximum values of benefit criteria and minimum values of cost criteria; and the negative ideal solution was determined from the minimum values of benefit criteria and maximum values of cost criteria after normalization and weighting. The Euclidean distance of each year from these two reference points was then calculated as:

$$D_i^+ = \sqrt{\sum_{j=1}^{n} \left(v_{ij} - v_j^+\right)^2} \quad (10)$$

$$D_i^- = \sqrt{\sum_{j=1}^{n} \left(v_{ij} - v_j^-\right)^2} \quad (11)$$

Based on these distances, the relative closeness coefficient (Ci) was calculated each year [25].

$$C_I = \frac{D_i^-}{D_i^+ + D_i^-} \quad (12)$$

The relative closeness coefficient (C_i) is between 0 and 1 [25]. A higher value indicates greater proximity to the positive ideal solution and therefore more favorable water-

availability conditions and lower water stress, whereas lower values indicate greater proximity to the negative ideal solution and higher water stress.

2.5. Water Stress Assessment Based on PROMETHEE

The annual water stress conditions in Al Hoceima Province were ranked using the multi-criteria decision-making method, the Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE II), as a complement to other methods. PROMETHEE is an outranking method based on pairwise comparisons, which is particularly well-suited to ranking a set of alternatives based on multiple criteria [10,19]. Some of these criteria may be conflicting. The method calls for two basic inputs: the weights of the criteria and a decision-preference function that represents the degree of preference one alternative has over the other for each criterion [10,19]. The criteria weights for this study were derived from the AHP analysis, and the same five criteria in the TOPSIS analysis (precipitation, temperature, evaporation, runoff, and population) were used in the PROMETHEE.

A global preference index for each pair of years was obtained by aggregating, across all criteria [10,19], the weighted preference functions for that pair.

$$\pi(a,b) = \sum_{j=1}^{n} \omega_j P_j(a,b) \quad (13)$$

where ω_j is the weight of criterion j, and $P_j(a,b)$ is the preference function associated with criterion j.

The positive outranking flow φ^+ and the negative outranking flow φ^- were then computed to represent, respectively, the degree to which a year outranks the remaining years and the degree to which it is outranked by them [10,19]. The net outranking flow was calculated as follows [10]:

$$\varphi = \varphi^+ - \varphi^- \quad (14)$$

where φ is the net flow, φ^+ is the positive outranking flow. and φ^- is the negative outranking flow. Higher positive net flow values indicate relatively favorable water availability, whereas lower negative net flow values indicate more severe water stress [10].

PROMETHEE II was chosen since it offers a complete ranking of all alternatives on the basis of their net flow values, whereas PROMETHEE I does not necessarily define a complete ranking of any alternatives, which are incomparable [10]. The annual rankings were then compared to the TOPSIS results to assess the consistency and robustness of the proposed multi-criteria assessment framework.

2.6. Innovative Trend Analysis (ITA)

The temporal trend of the annual WSI was first evaluated using linear regression. To complement the linear trend analysis, the non-parametric Mann–Kendall (MK) test was applied to assess the direction and statistical significance of the monotonic trend [26]. Sen's slope estimator was used to quantify the magnitude of the trend, together with its 95% confidence interval [27]. Serial dependence was also examined using the lag-1 autocorrelation coefficient and the Ljung–Box test [28]. The Hamed–Rao modified Mann–Kendall test was additionally applied to account for potential serial correlation [29]. We adopted a significance level of 0.05.

The annual TOPSIS and PROMETHEE scores were analyzed using the Innovative Trend Analysis (ITA) method presented in [11]. To obtain two sub-series of equal length, the first year (1981) was excluded from the ITA. The time series was divided into two equal sub-series representing the first and second halves of the study period. Each sub-series was sorted in ascending order, and the ordered values were plotted against each other on a

Cartesian graph. The trend was visually assessed by examining the distribution of points relative to the 1:1 line. Points located above the 1:1 line indicate an increasing trend, whereas points below the line indicate a decreasing trend. Points scattered around the 1:1 line indicate the absence of a clear trend. All calculations were performed using Microsoft Excel for Microsoft 365 MSO (Version 2610, Build 16.0.20527.20032, 64-bit; Microsoft Corporation, Redmond, WA, USA).

2.7. Comparative Analysis of TOPSIS and PROMETHEE Results

The annual water stress scores from TOPSIS and PROMETHEE were then examined using complementary graphical and statistical approaches to compare the performance of the two multi-criteria decision-making (MCDM) methods. Comparative line plots were used to investigate whether interannual patterns of water stress over the study period (1981–2025) were captured by both methods, beginning with the temporal evolution of the annual scores.

Second, the Pearson correlation coefficient (r) was calculated to assess the relationship between the annual TOPSIS and the annual PROMETHEE, and a scatter plot with a fitted linear regression was used to visualize it. A coefficient of determination (R^2) was also determined to measure the amount of variance accounted for by the linear relationship. Pearson's correlation was chosen because the data generated by TOPSIS and PROMETHEE were continuous numerical variables, and the objective was to assess the linear association between the two variables in terms of its strength and direction. Pearson's correlation coefficient is a suitable statistic for testing whether there is a linear association between two continuous variables [30]. It should be noted that correlation is a measure of the linear association between variables, and is not a measure of the agreement or interchangeability between methods [30]. Because the main output of both TOPSIS and PROMETHEE is the ranking of the study years, Spearman's rank correlation coefficient (ρ) was also calculated to assess the association between the rankings produced by the two methods.

Figure 1 gives an overview of the key steps of the methodological framework to be used in this research.

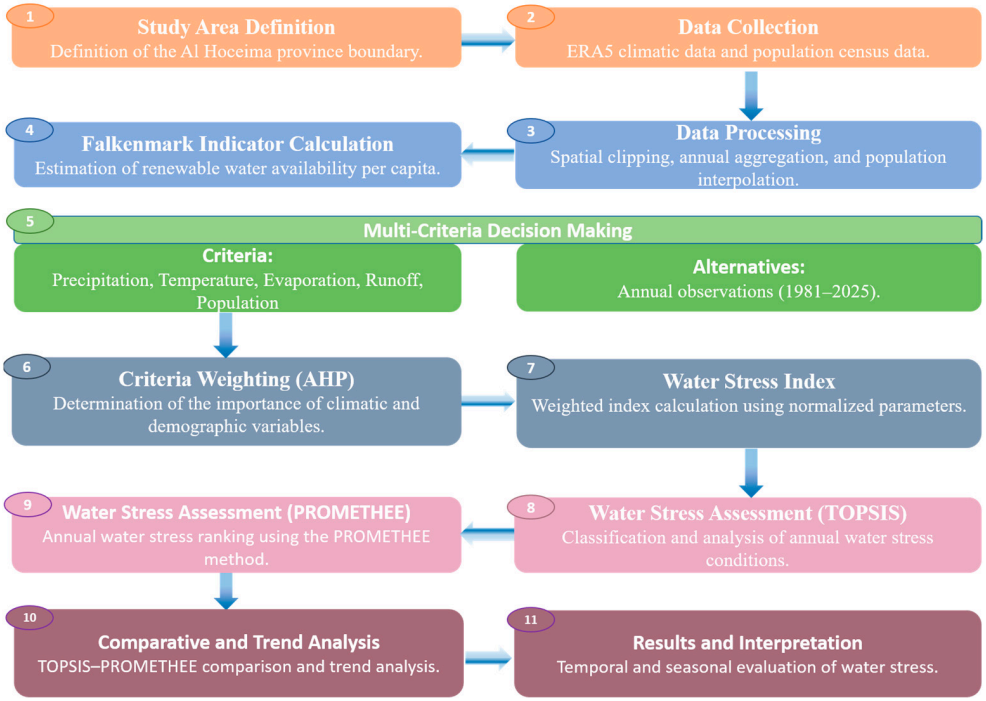


Figure 1. Methodological framework for water stress assessment.

3. Case Study

This study focuses on Al Hoceima Province in northern Morocco, which is part of the administrative region of Tanger-Tetouan-Al Hoceima [31]. The climate of the area is Mediterranean, characterized by hot, dry summers and mild, wet winters [32]. Rainfall in the area exhibits strong spatial and interannual variability. To some extent, this variability depends on the geomorphological environment of Al Hoceima Province, as the Rif Mountains and the Mediterranean coastline provide distinct climatic conditions in the region [33]. Precipitation is predominantly observed between October and March, and the highest rainfall occurs mostly between December and February [33]. In turn, rising water shortages and human pressures can affect human activities and natural ecosystems, increasing stress on water resources [34]. Figure 2 shows the administrative boundaries of Al Hoceima Province and its communes.

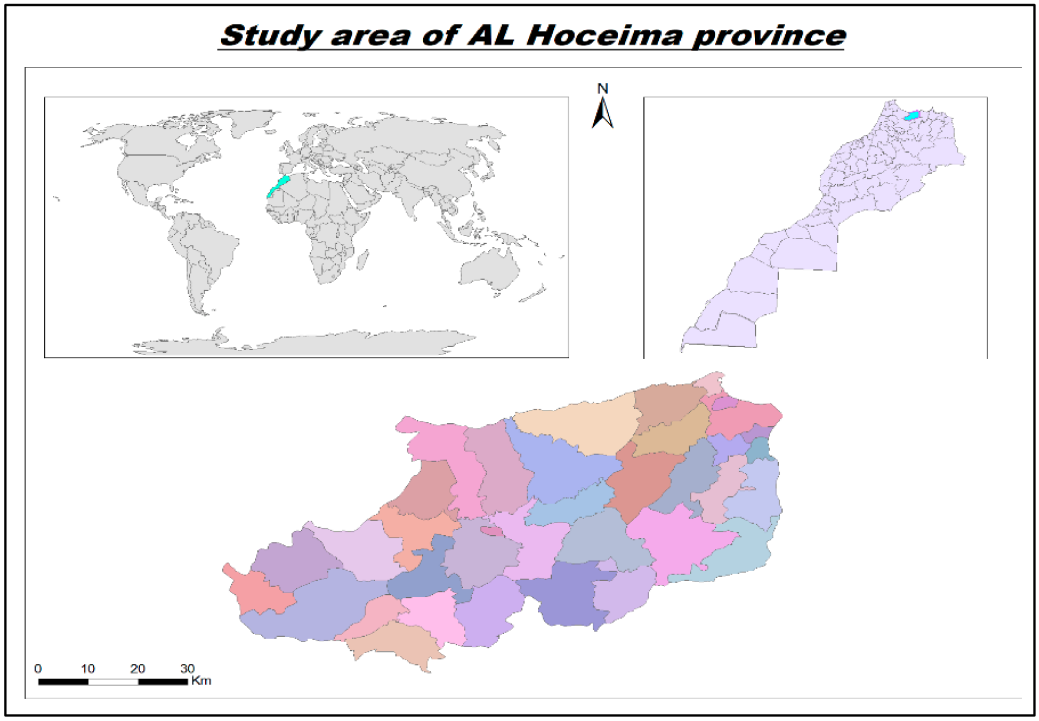


Figure 2. Location of the study area (Al Hoceima Province, Morocco) [35].

The administrative boundary of the province, which lies at the second administrative level, was selected as the study area according to the Global Administrative Areas (GADM) database v4.1 [35]. The spatial data were provided in geographic coordinates using the WGS84 datum. The resulting shapefile was further manipulated in Python 3.14 (64-bit; Python Software Foundation) using the GeoPandas library to extract the polygon corresponding to Al Hoceima Province. This polygon was used to spatially clip the ERA5 datasets, retaining only grid cells within the provincial boundary for the analysis. The climatic and hydrological data were then clipped to this geometry, retaining only data pertinent to the study region.

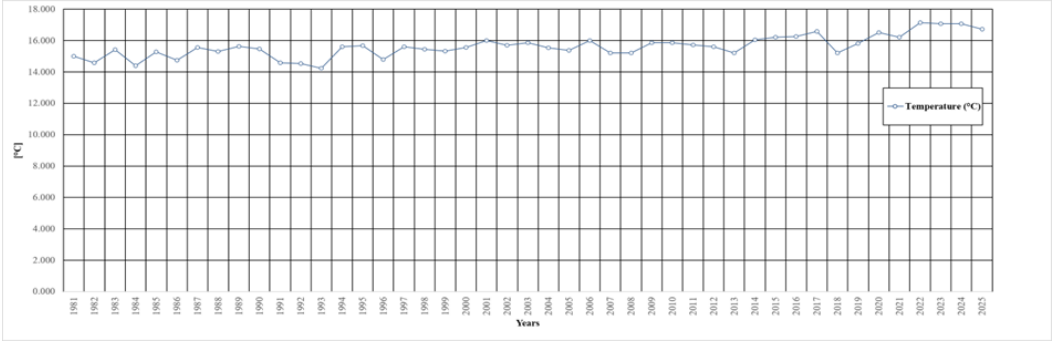
3.1. Hydro-Climatic Data

This study used four variables, namely 2-m air temperature, total precipitation, evaporation, and runoff. Water scarcity occurs when water demand approaches or exceeds available supply. In this study, however, direct water-use data were not available, and water stress was therefore assessed using hydro-climatic and demographic indicators as proxies of pressure on water resources. The primary natural source of freshwater is precipi-

tation, which increases recharge during wet years and is reduced during dry years, putting pressure on water resources. Temperature is also a significant factor, as rising temperatures increase the rate of evapotranspiration and water loss through soils, reservoirs, and plants. Evaporation reduces the volume of water available for human activities and ecosystems. Runoff represents the proportion of precipitation that supplies rivers and streams and is regarded as an effective measure of renewable water supply. The Copernicus Climate Data Store was accessed to extract monthly data for the period 1981–2025, and the data were manipulated using Python-based libraries such as Xarray, GeoPandas, and rioxarray. The resulting annual values of the variables are presented in Figure 3.



(**a**)



(**b**)

Figure 3. Annual climate and hydrological variables in Al Hoceima Province (1981–2025): (**a**) precipitation, runoff, and evaporation; (**b**) 2-m air temperature.

Since ERA5 data are available on a regular spatial grid, spatial aggregation was needed to obtain representative values for a specific geographical unit. This procedure is generally used in climatic analysis to convert grid-cell data to regional values. Accordingly, all grid cells that fell within the provincial boundary were clipped and spatially averaged to create one time series representative of Al Hoceima Province [36,37].

In the long-term analysis, annual time series were created out of the monthly values. Annual water fluxes were calculated as the sum of precipitation, evaporation, and runoff, while temperature was averaged over the year to characterize the climatic conditions associated with atmospheric water loss and water stress [38]. To make all datasets homogeneous, all variables were converted to standard climatological units (°C and mm). The proxy for renewable water resources was runoff, since the availability of renewable freshwater is tightly connected to long-term runoff and river discharge in hydrological systems [39]. Temperature was expressed in °C, while precipitation, evaporation, and runoff were expressed in mm.

3.2. Population Data and Demographic Interpolation

The official Moroccan population censuses conducted by the High Commission for Planning [40–42] were consulted to obtain the population data. In places where water is

already a scarce resource, population can rapidly escalate demand for freshwater. This is why population dynamics are often part of water stress assessments to reflect demographic pressure on available water resources. Census records were available for the years 1994, 2004, 2014, and 2024.

To create a continuous dataset for the entire study period (1981–2025), population values between census years were interpolated using an exponential function. This method was also used to estimate population values before 1994 due to a lack of homogeneous data aligned with current administrative boundaries. This method assumes that the growth rate of the census observations is constant [43]. In a particular year, the population was estimated as:

$$P_t = P_0 \times e^{r\,(t-t0)} \quad (15)$$

P_t represents the population at year t, P_0 is the population at the nearest census year t_0, and r represents the annual growth rate.

To maintain consistency with the climatic variables, the estimated annual population values were assigned to all months of the same year. This ensures temporal alignment between the demographic and climatic datasets used in the analysis.

4. Results

4.1. Relative Importance of Water Stress Criteria

The Analytic Hierarchy Process was used to calculate the weights of the chosen criteria. The consistency of the pairwise comparison matrix was confirmed by computing the maximum eigenvalue, consistency index (CI), and consistency ratio (CR). The calculated consistency ratio (CR = 0.0635) is below the recommended threshold of 0.1, indicating that the pairwise comparison matrix is consistent and that the derived weights are reliable.

The weights obtained suggest that runoff is the dominant factor affecting water stress conditions in the study area, accounting for 48.81% of the total weight of importance. The AHP-derived weights and the consistency statistics are presented in Table 5. The result indicates that surface water availability plays a major role in controlling variations in water availability. Precipitation is the second-ranked parameter (24.00%), highlighting its vital importance for water resources, streamflow, and groundwater recharge. Evaporation is a major source of water loss, accounting for 13.59%, and it reduces the utilization of water resources, especially in arid climates. Temperature contributes 8.97% and primarily influences water stress indirectly by increasing evapotranspiration and water loss. The smallest weight (4.63%) was assigned to population, indicating that water stress in the study area is driven not only by population pressure but also by variability in hydroclimatic conditions.

Table 5. AHP weights and consistency values.

Criteria	Weight	λ_{max}	CI	RI	CR
Runoff	0.4881	5.2845	0.0711	1.12	0.0635
Precipitation	0.2400				
Evaporation	0.1359				
Temperature	0.0897				
Population	0.0463				

Overall, the results highlight the predominance of natural hydrological and climatic factors in determining water stress conditions in Al Hoceima Province.

The sensitivity analysis showed that the relative importance of the five criteria remained consistent under the tested weight perturbations. The ranking of the criteria remained unchanged across the tested perturbation scenarios, although the numerical weights varied. The complete sensitivity results are presented in Table 6.

Table 6. Sensitivity analysis of AHP-derived weights under ±10%, ±20%, and ±30% perturbations.

Criterion Perturbed	−30%	−20%	−10%	10%	20%	30%
Runoff	0.4004	0.4329	0.4620	0.5121	0.5338	0.5537
Precipitation	0.1811	0.2017	0.2213	0.2578	0.2749	0.2911
Evaporation	0.0991	0.1117	0.1239	0.1474	0.1587	0.1697
Temperature	0.0644	0.0751	0.0825	0.0964	0.1029	0.1134
Population	0.0329	0.0392	0.0428	0.0496	0.0527	0.0593

The results indicate that moderate changes in the initial AHP weights did not alter the overall relative ordering of the criteria.

More importantly, the recalculated WSI values under the 30 sensitivity scenarios preserved the same overall temporal evolution. The same main stressful and less stressful years were consistently identified across all tested scenarios, although the WSI values varied in magnitude. These results indicate that the overall temporal assessment of water stress is robust to the tested perturbations of the AHP weights. The complete WSI results for the 30 sensitivity scenarios are provided in Supplementary Table S1.

4.2. Temporal Change in Al Hoceima Water Stress Index

Figure 4 shows the time series of the Water Stress Index (WSI) in Al Hoceima. There is considerable variability in the WSI over time, with the lowest value of 0.218 in 2010 and the highest value of 0.829 in 2017. The WSI also reached a very high value in 2023 (0.824), among the highest values recorded during the study period. Despite this variability, most years show relatively elevated WSI values, indicating that water stress is a persistent and recurrent issue in the region. Only a few years experienced relatively favorable conditions with lower levels of water stress, reflecting temporary improvements in water availability rather than a long-term recovery of water resources.

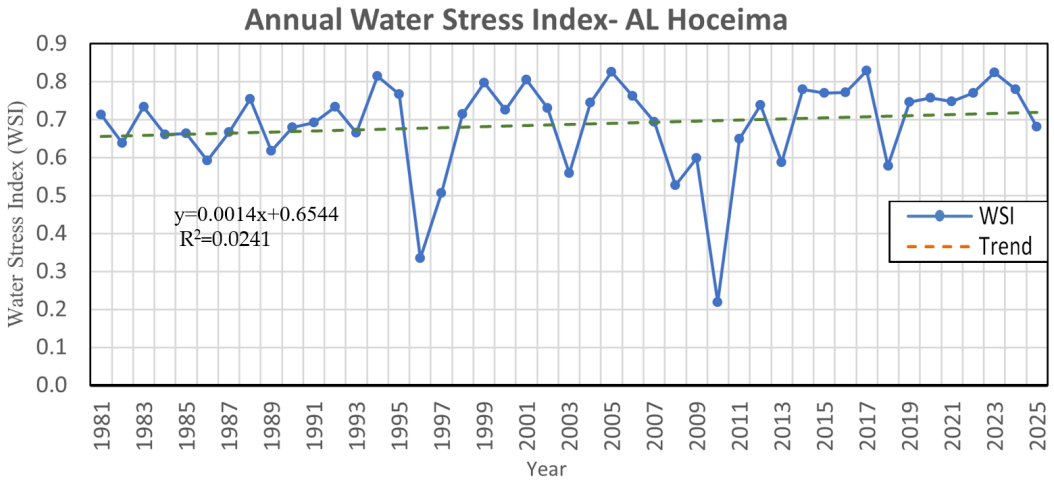


Figure 4. Temporal variability of the Water Stress Index (WSI) in Al Hoceima Province (1981–2025).

The index also shows periods of more pronounced water stress, especially in the mid-1990s, the latter half of the 2000s, and again in the early 2020s. These peaks are associated

with high hydrological variability, during which per capita water availability may decrease and pressure on local water resources becomes more severe. This is clearly observed in 2010, when the WSI reached its lowest value (0.218), likely reflecting favorable hydro-climatic conditions with higher rainfall and runoff and, consequently, greater water availability. In contrast, the most unfavorable conditions were recorded in 2017, when the WSI reached its highest value (0.829). Water stress was also very high in 2023 (0.824), consistent with the severe drought conditions reported nationally in Morocco. These high values may reflect the combined effects of prolonged dry conditions, rising temperatures, increased evaporation, and growing pressure on available water resources.

Overall, the linear regression showed a positive slope of 0.00144 WSI/year, but the trend was not statistically significant (R^2 = 0.0241, p = 0.309). The Mann–Kendall test indicated a statistically significant increasing monotonic trend (Kendall's tau = 0.224, p = 0.031). Sen's slope was estimated at 0.00208 WSI/year, with a 95% confidence interval of 0.00017–0.00392 WSI/year. The lag-1 autocorrelation was 0.234, while the Ljung–Box test did not indicate significant serial correlation (p = 0.105). The Hamed–Rao modified Mann–Kendall test also indicated a statistically significant increasing trend (Kendall's tau = 0.224, p = 0.0085).

The WSI results were further compared with officially documented precipitation conditions in Morocco. According to Maroc Météo, 2023 recorded the largest national precipitation deficit since at least 1980, reaching approximately 48%, followed by 2017 with a deficit of about 44% [44]. In the present study, the highest WSI value was recorded in 2017 (WSI = 0.829), while 2023 also showed a very high WSI value (0.824). The WSI was also high during the subsequent dry period, with values of 0.757, 0.747, 0.769, and 0.779 in 2020, 2021, 2022, and 2024, respectively. These results are consistent with the persistent dry conditions reported nationally by Maroc Météo. In particular, Maroc Météo reported that the period 2019–2023 was the longest and most severe sequence of consecutive dry years, while 2024 marked the sixth consecutive year of drought [45]. This consistency between the WSI results and officially documented dry conditions provides additional support for the ability of the WSI to capture periods of elevated water stress.

In general, wet years reduce the WSI, whereas dry and hot years increase it due to reduced water availability and intensified hydro-climatic stress. These observations highlight the increasing vulnerability of water resources in Al Hoceima Province and emphasize the strong influence of hydro-climatic variability and human pressure on water stress conditions over time.

4.3. Water Stress Drivers: Role of Hydro-Climatic and Socio-Economic Factors

Figure 5 illustrates the contribution percentages of the different variables to the Water Stress Index (WSI) in Al Hoceima between 1981 and 2025.

The findings clearly indicate that the most important aspect of the index calculation is the runoff. This does not mean that runoff increases water scarcity, but it indicates a strong negative correlation between runoff and water stress. Water scarcity is greater when runoff is lower, as there is less water in rivers and reservoirs, which directly influences the region's water balance.

Another important factor is precipitation, particularly during years of high climate variability. Precipitation regulates the recharge of both surface and groundwater systems, and thus any alterations in precipitation are readily reflected in water availability. Its impact, however, is not independent, since it relies heavily on how much rainfall becomes runoff. This is one of the main aspects of precipitation and runoff interactions, which is important in interpreting water stress changes over time. The role of temperature and evaporation is moderate, yet they still play a significant role in semi-arid climates such as Al Hoceima. An

increase in temperature enhances evaporation rates, resulting in increased water loss in soil, reservoirs, and open water bodies. In long, hot, and dry seasons, this process reduces the effective amount of available water and increases the overall water stress.

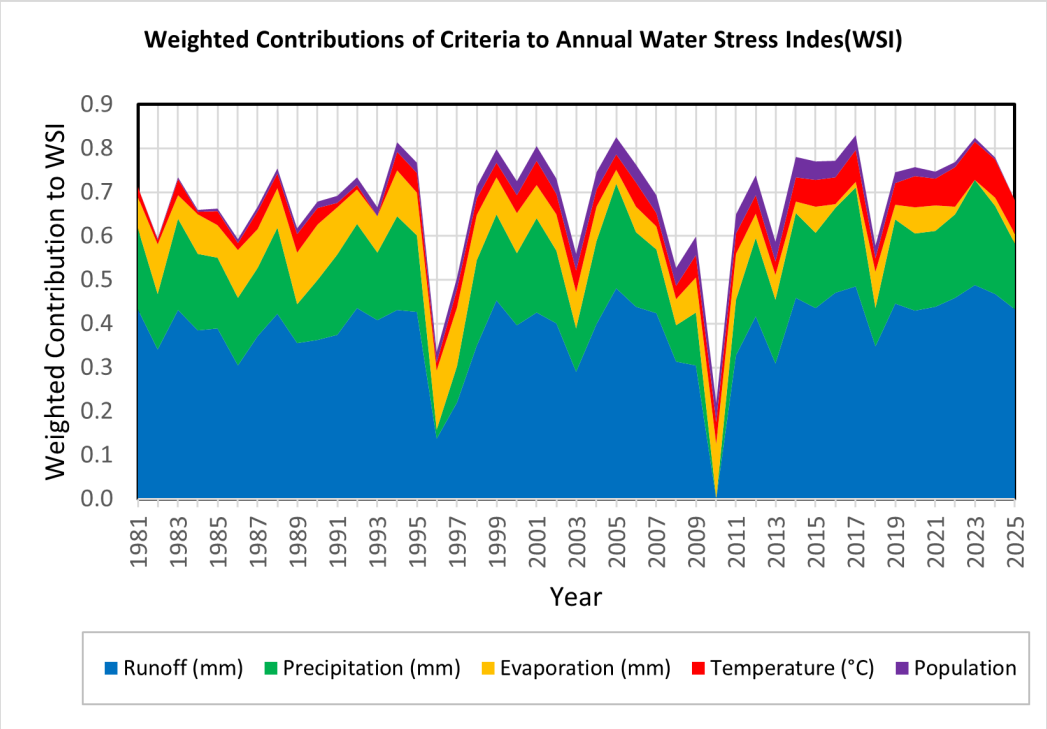


Figure 5. Relative contribution of hydro-climatic and demographic variables to the Water Stress Index (WSI).

Population contributes the least among the other factors, although the impact is still felt. Demographic pressure can still contribute to water stress because population growth increases the number of people potentially dependent on the available water resources, which may exacerbate pressure on water resources in dry years when supply is already constrained. All in all, the findings point to the fact that hydrological and climatic factors, particularly runoff and precipitation, dominate water scarcity in Al Hoceima, with socio-economic factors playing a minor but increasing role.

4.4. Monthly Variability of Water Stress Conditions

Figure 6 shows the changes in the monthly values of the Water Stress Index (WSI) in Al Hoceima and makes it clear that there is a strong seasonal trend largely influenced by climatic factors. The WSI increases from spring onwards, reaching its maximum in July (0.868), then shows very high values in August (0.853) and June (0.838). This period is characterized by high temperatures and intense evaporation, which means that water supply is greatly reduced. Simultaneously, rainfall is extremely low, and runoff is reduced, increasing water stress and diminishing water availability to an even greater extent. Conversely, the lowest WSI values are in winter, particularly in January and December, both with a WSI of approximately 0.662. These reduced values demonstrate more desirable hydrological environments, as winter precipitation is a source of recharge to the surface waters and even sustains groundwater replenishment. Winter precipitation increases runoff, enhancing water availability in general and decreasing the level of scarcity in the region.

The months between these two extremes represent a gradual transition between these conditions. In spring (March–May), the WSI rises continuously from 0.678 to 0.764, indicating increasingly stressful hydro-climatic conditions as evaporation rises and water

availability becomes more limited. The opposite process occurs during autumn (September–November), when the index steadily declines to 0.675 due to falling temperatures and gradually increasing rainfall, which improves hydrological conditions.

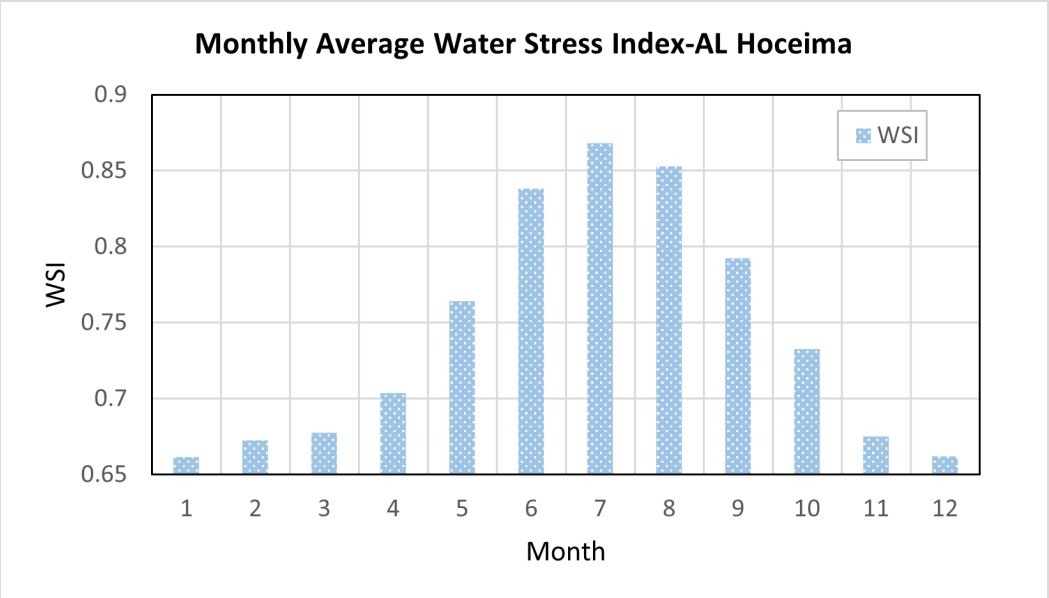


Figure 6. Monthly variation in the Water Stress Index (WSI) in Al Hoceima Province.

The findings generally indicate that seasonal climate variability plays a significant role in controlling water scarcity in Al Hoceima. Dry summers and wet winters form an evident annual water-stress cycle, common to Mediterranean areas.

4.5. Relationship Between WSI and Falkenmark Water Stress Index

Figure 7 shows a strong negative relationship between the Water Stress Index (WSI) and the Falkenmark Index during the study period. Spearman's rank correlation confirmed this relationship, with $\rho = -0.912$ ($p < 0.001$). This inverse relationship is expected because increases in the Falkenmark Index imply more water resources per capita, which is typically associated with reduced water stress and, therefore, lower WSI values. Conversely, when the Falkenmark Index declines, per capita water availability is lower, and WSI tends to rise, indicating increased water scarcity.

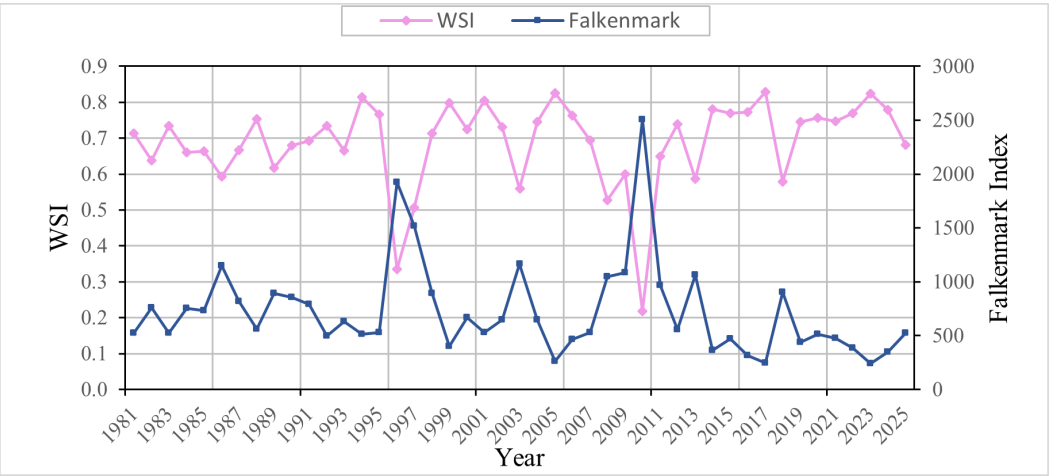


Figure 7. Relationship between Water Stress Index (WSI) and Falkenmark indicator.

Nonetheless, the correlation is not exactly proportional, indicating that the two indices do not capture precisely the same factors of the water scarcity problem. The Falkenmark Index is primarily based on per capita water supply, whereas the WSI is a composite of a number of hydrological and climatic variables including runoff, precipitation, temperature, and evaporation. Due to this difference, there are some divergences between the two indicators in some years, particularly where climatic conditions have a major impact on water resources that are not well reflected in population-based indicators.

A good illustration is the period around 2010, when the WSI declines significantly and the Falkenmark Index reaches its highest value. This likely reflects a wet year with better water availability, leading to lower water stress. This also highlights the sensitivity of the two indicators to annual variations in hydro-climatic conditions.

The Falkenmark Index exhibits high fluctuations over the long term, with certain periods of low values that could be attributed to population increase as well as variations in climate. Simultaneously, the WSI remains relatively high throughout most of the study period, indicating that water stress is a long-term problem in the area despite yearly variations. Overall, the comparison shows that both indices capture the general variations in water stress, although they respond differently to the underlying factors. The WSI is more responsive to hydro-climatic conditions, whereas the Falkenmark Index is more directly influenced by population pressure and per-capita water availability. The combination of the two provides a more detailed picture of the dynamics of water stress in Al Hoceima.

4.6. TOPSIS-Based Water Stress Assessment

4.6.1. Ranking of the Most Stressful Years Based on TOPSIS

Table 7 shows the ten years showing the most water stress based on the TOPSIS ranking. These years have the lowest Ci values and are thus the most critical years with regard to water availability. The year 2005 recorded the lowest score, followed by 2017 and 2023, indicating particularly stressful conditions. These very low values indicate dry conditions, including reduced precipitation and runoff, as well as increased evaporative losses, in addition to negative hydro-climatic conditions. An interesting aspect of the ranking is that recent years are heavily represented. Among the ten years with the lowest TOPSIS scores, 7 have occurred since 2014 (2014, 2016, 2017, 2019, 2022, 2023 and 2024), suggesting worsening water stress over the past decade.

Table 7. Top 10 high water stress years based on TOPSIS results.

Rank	CI	Year	Stress Level
1	0.051	2005	Most stressful
2	0.058	2017	Most stressful
3	0.062	2023	Most stressful
4	0.075	2024	Most stressful
5	0.078	2016	Most stressful
6	0.083	1999	Most stressful
7	0.087	2014	Most stressful
8	0.090	2022	Most stressful
9	0.106	2019	Most stressful
10	0.117	1992	Most stressful

The fact that 2022, 2023, and 2024 have all fallen within the top three most critical years since 2011 reinforces ongoing drought conditions and indicates declining water

availability in the region. The older years (1992 and 1999) indicate that high water stress is not a fresh occurrence in the study area. However, the clustering of extreme events in more recent decades may be due to a combination of increased climatic variability, periodic droughts, and increasing stress on available water resources. Overall, the ranking shows that these episodes of extreme water stress are occurring at regular frequencies and highlights the vulnerability of Al Hoceima Province with respect to hydrological variability and water scarcity.

4.6.2. Year-to-Year Variability of Water Stress Conditions Based on TOPSIS

The interannual variability of the TOPSIS-based Water Stress Index in Al Hoceima Province from 1981 to 2025 is represented in Figure 8. The long-term mean value (C_i = 0.222) is used as a reference to distinguish relatively more favorable and less favorable water-availability conditions. The results show significant annual fluctuations, with most annual values below the long-term mean, suggesting that water stress dominates during the time series. A small number of years had above-average conditions, indicating short-term gains in water supply. The most favorable hydrological conditions for both periods are observed in 1996 (Ci = 0.72) and particularly in 2010 (Ci = 0.94), which are two exceptional peaks. The high scores are probably related to very wet years with more precipitation and runoff to relieve water stress. The high values were also observed in 1986, 1989, 1997, 2003, 2008, 2009, 2011, 2013, and 2018, although these favorable conditions were temporary and did not follow a clear temporal pattern. These years are irregularly distributed over time, indicating that water availability is more likely to improve temporarily due to short-term hydro-climatic fluctuations than as a result of a long-term recovery of water resources. On the other hand, a few years were characterized by very low values; for instance, 2005, 2017, 2023, and 2024 were among the most stressful years according to the TOPSIS ranking. The trend of lower scores after 2014, along with the successive stressful years from 2022 to 2024, indicates increasing persistence of drought conditions and a decrease in water availability in recent decades. Overall, the dominance of years with values below the long-term mean indicates that water scarcity is a frequent and structural feature of Al Hoceima Province. These fluctuations further emphasize the region's reliance on precipitation and runoff variability, as well as its susceptibility to extended periods of low rainfall.

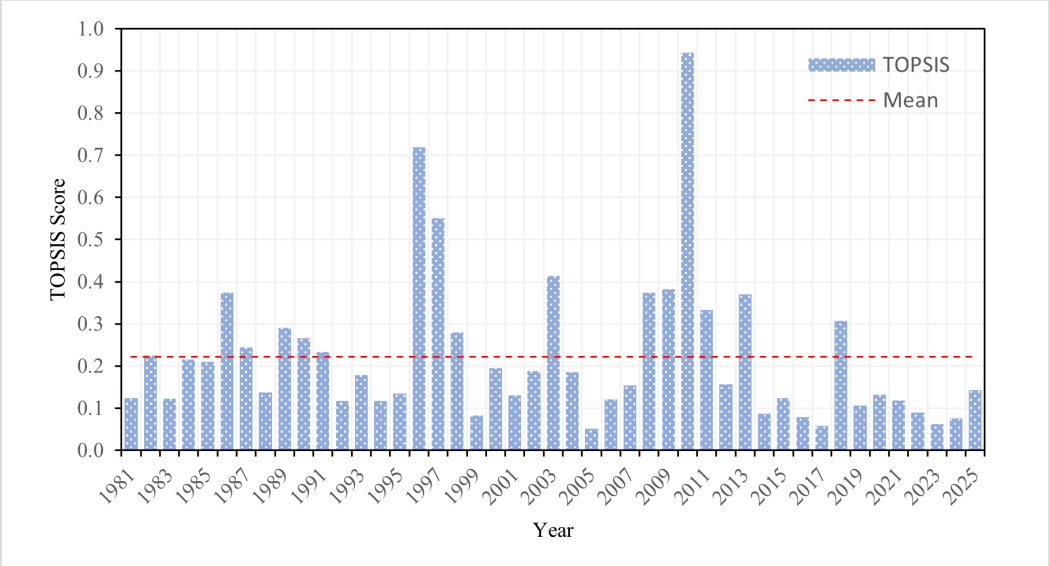


Figure 8. Interannual variability of TOPSIS in Al Hoceima Province (1981–2025).

4.6.3. Innovative Trend Analysis of Water Stress Conditions Based on TOPSIS

The Innovative Trend Analysis (ITA) of the TOPSIS performance scores is shown in Figure 9. To meet the ITA requirement of dividing the time series into two subseries of equal length, one year was removed from the original time series of 45 observations. The earliest year (1981) was not analyzed to keep only the most recent observations and give a more representative water stress assessment. The scatter points display a heterogeneous distribution with respect to the 1:1 reference line. Most low TOPSIS scores cluster around the origin and fall below the reference line, suggesting that values in the second half of the study period are generally lower than in the first half. As TOPSIS scores decrease, water stress increases, indicating that water availability is declining and water stress is rising over time. The points with TOPSIS values in the moderate range are scattered around the 1:1 line, with some above and some below. This means that there is no significant trend for the intermediate range of TOPSIS scores. Short-term changes in water availability may have been present, but not enough to alter the general hydrological response of the area.

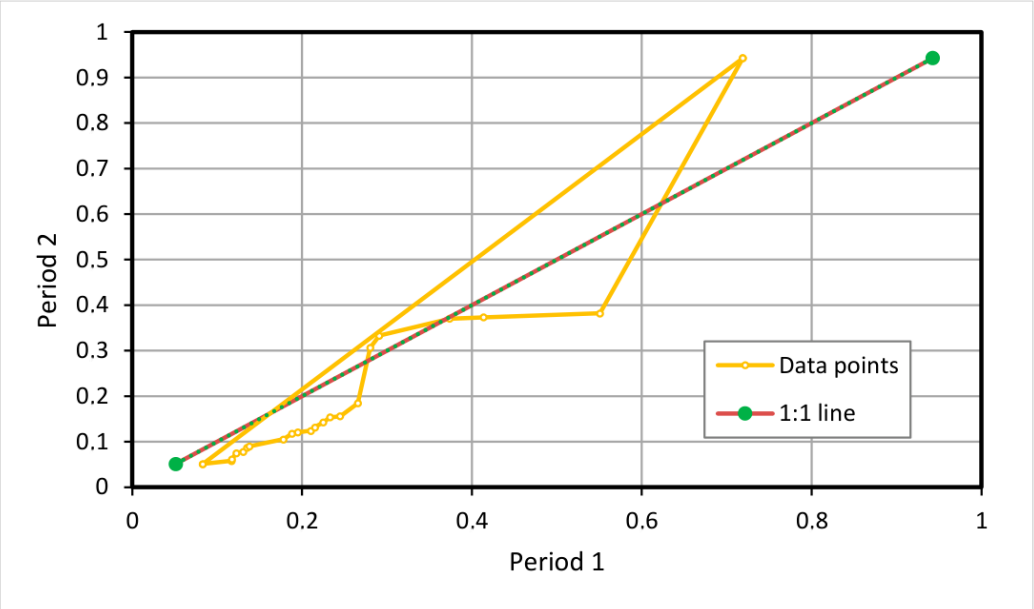


Figure 9. Innovative Trend Analysis of TOPSIS Scores in Al Hoceima Province (1981–2025).

There is one distinct point well above the 1:1 reference line that corresponds to the pair of exceptionally high TOPSIS scores in the first and second subperiods, respectively, in 1996 (0.718) and 2010 (0.943). This suggests that water availability conditions in both subperiods were desirable, and TOPSIS scores were higher in the second subperiod than in the first. This point indicates that the maximum water availability conditions were more favorable in the second subperiod than in the first, and highlights two exceptionally wet years with high precipitation and increased runoff. These points, however, are clearly distinct from most observations and should not be considered typical of the long-term hydrological regime.

The overall results of the ITA indicate increasing water stress in Al Hoceima Province over the past decades, showing inhomogeneity and anisotropy of the obtained data. Breaks and crossings of the polygon values over the 1:1 line undoubtedly indicate variability of the data. The majority of the lower TOPSIS scores below the 1:1 line show that overall water availability is decreasing, while only two isolated years with high TOPSIS scores show that there are only several short periods of abundant water resources, which are not enough and do not compensate for the long-term trend of water resources becoming scarce.

4.7. PROMETHEE-Based Water Stress Assessment

4.7.1. Ranking of the Most Stressful Years Based on PROMETHEE

The ten most stressful years were identified using the PROMETHEE ranking, based on the lowest net-flow values; the ranking is given in Table 8. The years with the lowest net flow values are the most critical in terms of water availability. The year 1994 recorded the lowest score, followed by 2023 and 2017, indicating particularly stressful conditions. The strongly negative values indicate less favorable hydro-climatic conditions, characterized by low precipitation, low runoff, and high evaporative loss. One special aspect of the ranking is that both past and recent extreme events are included. In recent decades, three of the ten most water-stressed years occurred after 2010 (2017, 2023, and 2024), indicating that severe water stress has persisted. In particular, the years 2023 and 2024 were among the most critical, indicating that the region experienced constrained water availability and drought conditions during these recent years. Earlier years (1983, 1992, 1994, 1999 and 2001) suggest that severe water stress has been an ongoing occurrence at the study site for many decades and is not a recent phenomenon.

Table 8. Top 10 most stressful years based on PROMETHEE results.

Rank	CI	Year	Stress Level
1	−0.662	1994	Most stressful
2	−0.636	2023	Most stressful
3	−0.572	2017	Most stressful
4	−0.567	2005	Most stressful
5	−0.551	1999	Most stressful
6	−0.516	1992	Most stressful
7	−0.490	2001	Most stressful
8	−0.430	2006	Most stressful
9	−0.416	2024	Most stressful
10	−0.411	1983	Most stressful

Multiple extreme events in recent years, however, may be due to the cumulative impacts of hydro-climatic variability and repeated drought events. In general, the PROMETHEE ranking highlights the vulnerability of Al Hoceima Province to water scarcity and shows that episodes of severe water stress are frequent and long-lasting.

4.7.2. Year-to-Year Variability of Water Stress Conditions Based on PROMETHEE II

The results of water stress conditions in Al Hoceima Province from 1981 to 2025 are shown in Figure 10, which is based on the PROMETHEE net flow. Water availability conditions are considered relatively good if positive and relatively poor if negative. The findings show significant inter-annual variability over the years, which underlines the high temporal variability of regional water resources. Several years have positive net flow, representing relatively favorable hydrological conditions. The highest positive values are observed in 2013 (0.65) and 1996 (0.62), followed by 1986, 2009, 2010, 2011 and 2018, all of which show markedly positive scores. These years are probably related to wetter-than-normal conditions and higher runoff, which temporarily reduced water stress. These favorable conditions, however, are irregularly distributed and do not show any definite temporal pattern, indicating that improvements in water availability are more related to short-term hydro-climatic variability than to a long-term recovery of water resources. However, there are also several years with very low net flow values. The most critical conditions occur in

1994 (−0.66) and 2023 (−0.64), followed by 2005, 2017, 1999, 2001, and 2024, all of which present substantially negative scores. These years may correspond to drought periods characterized by reduced precipitation and runoff, which placed greater pressure on available water resources. While both wet and dry years are observed throughout the study period, there is an increasing number of negative values in recent decades, particularly during the period 2022–2024. The pattern indicates that drought conditions have been getting progressively worse and water availability has been declining in recent years. The positive and negative net flows confirm the high sensitivity of water resources in Al Hoceima Province to climatic variations.

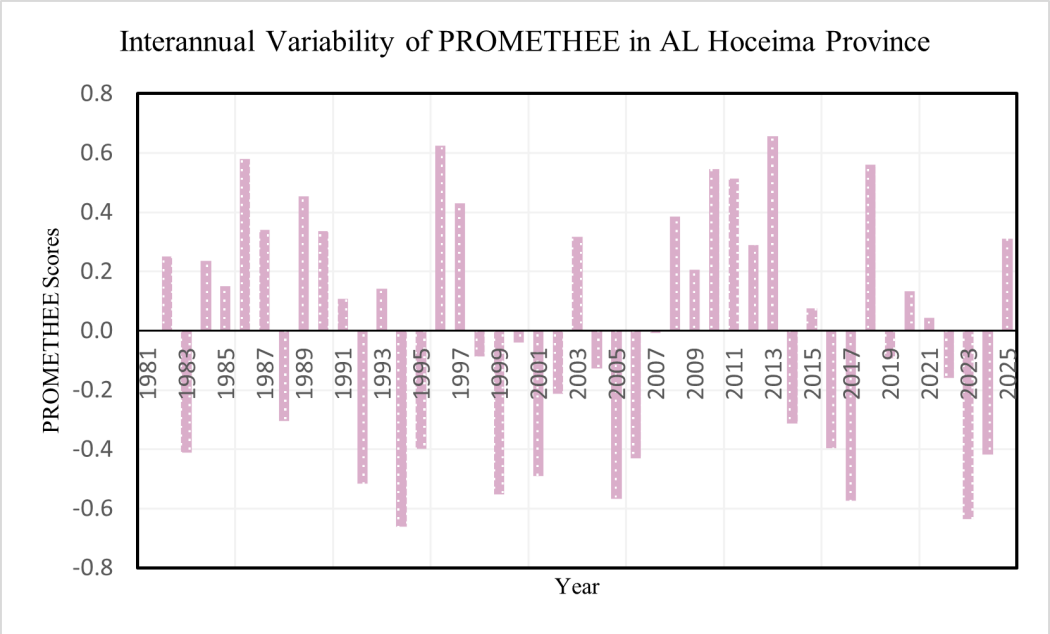


Figure 10. Interannual variability of PROMETHEE in Al Hoceima Province (1981–2025).

In general, the results of the PROMETHEE analysis corroborate that water availability in the province fluctuates considerably from year to year, with favorable conditions occurring only in isolated years, whereas severe water stress conditions occur more frequently. The predominance of negative net flow values in recent years further highlights the increasing influence of climatic variability on the sustainability of the region's water resources.

4.7.3. Innovative Trend Analysis of Water Stress Conditions Based on PROMETHEE

The Innovative Trend Analysis of the PROMETHEE performance scores is shown in Figure 11. To meet the ITA requirement of dividing the time series into two subperiods of equal length, one year was removed from the original time series. The earliest observation (1981) was omitted to allow a balanced comparison between the first and second halves of the study period, resulting in two subperiods of 22 observations each. The scatter points are grouped near the 1:1 reference line, which shows that the magnitude of the PROMETHEE performance scores is relatively similar between the two subperiods. However, the distribution is not completely symmetrical. Most lower performance scores lie slightly below the 1:1 line, suggesting that scores in the second half of the study period tend to be lower than those in the first half. Lower PROMETHEE net flow values indicate poorer water availability and higher water stress, suggesting a gradual deterioration of hydrological conditions over time. The intermediate performance scores are distributed on both sides of the reference line and show no clear trend. The scatter of the distribution implies that other controlling factors, such as interannual hydro-climatic variability, are at play, meaning that the observations do not correspond exclusively to either wet or dry conditions but rather

alternate between them. The highest scores are located above the 1:1 reference line, indicating that the maximum PROMETHEE score in the second subperiod was higher than that in the first subperiod. These correspond to very favorable hydrological years, mainly 1996 (0.62) and 2013 (0.65), with high precipitation and runoff, thereby temporarily improving water availability. Such favorable years, however, are isolated and do not represent the dominant hydrological conditions of the study area.

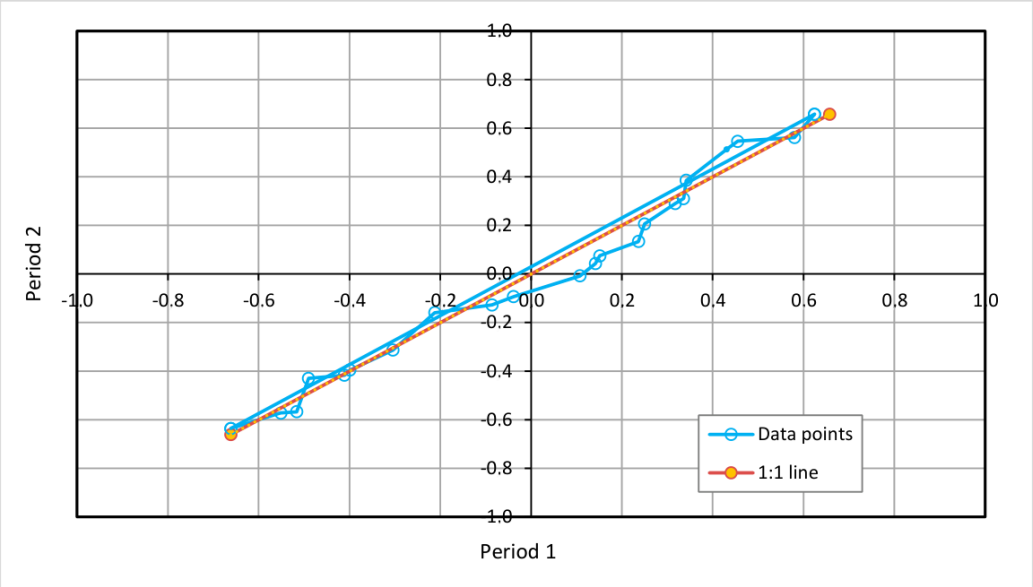


Figure 11. Innovative Trend Analysis of PROMETHEE Scores in Al Hoceima Province (1981–2025).

The overall ITA results show a slight decrease in PROMETHEE performance scores, particularly for low and moderate values, suggesting a tendency toward higher water stress in recent decades. On the other hand, frequent breaks and crossings of the value polygon over the 1:1 line indicate inhomogeneity and anisotropy of the calculated values. The years of good water availability remain isolated and cannot compensate for the general trend of water scarcity, making Al Hoceima Province increasingly vulnerable to frequent periods of water stress.

4.8. Comparative Analysis of MCDM Methods

4.8.1. Comparative Analysis of TOPSIS and PROMETHEE Temporal Trends

Figure 12 shows the comparison of the annual water stress scores calculated from both the TOPSIS and PROMETHEE methods for the period 1981–2025. Both methods display large interannual fluctuations in the water stress, with some years showing favorable hydrological conditions and others unfavorable conditions, as discussed above. The two methods are based on different multi-criteria decision-making principles and different scoring scales, but generally show the same temporal evolution. Both methods consistently identify the main wetter years, including 1996, 2003, 2008, 2010, 2013 and 2018, as well as the main dry years, such as 2005, 2017, 2023 and 2024. The similarity between the two methods provides a comparable picture of the annual water stress conditions. There are some differences, though, in the size of the annual scores. The PROMETHEE method shows more over-time variation, with lower negative scores, and the TOPSIS method shows higher positive scores in the very good hydrological years. This difference relates to methodological features of the two approaches. The differences between the alternatives are emphasized in PROMETHEE, while in TOPSIS each year is evaluated relative to the distance from the ideal and anti-ideal solutions, which provides smoother score variations.

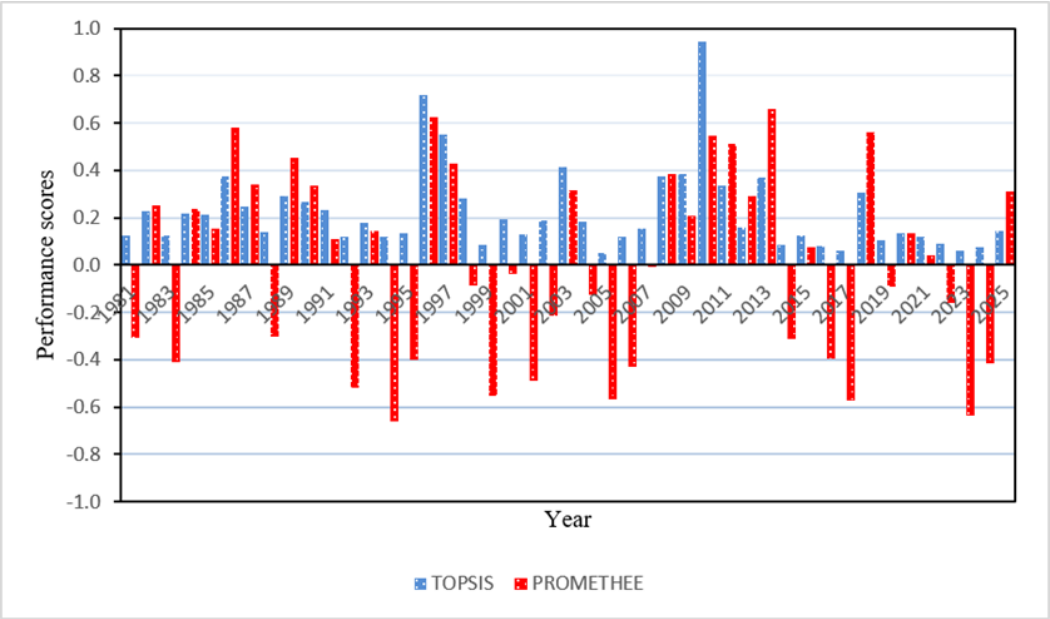


Figure 12. Comparison of Annual Water Stress Scores Obtained Using TOPSIS and PROMETHEE (1981–2025).

Both methods complement each other in evaluating water stress conditions. Their ability to identify the same critical wet and dry years shows that both methods capture similar temporal patterns, and PROMETHEE's higher sensitivity to unfavorable years provides a better understanding of hydrological extremes. Thus, the results from the two MCDM methods provide complementary information on water stress conditions in Al Hoceima Province.

4.8.2. Correlation Analysis Between TOPSIS and PROMETHEE Methods

Figure 13 presents the relationship between the annual water stress scores obtained using the TOPSIS and PROMETHEE methods. There is a positive linear relationship, indicating that years with higher TOPSIS scores tend to have higher PROMETHEE net flow values. The coefficient of determination (R^2 = 0.512) suggests that around 51.2% of the variation in the PROMETHEE scores can be explained by the TOPSIS scores. There is a positive correlation between the two methods (r = 0.72). Because the main output of both methods is the ranking of years, Spearman's rank correlation was also calculated. The Spearman rank correlation coefficient was ρ = 0.872, indicating a positive association between the rankings produced by the two methods. The results show that the mathematical bases of TOPSIS and PROMETHEE are somewhat different, but the two show a similar evaluation of annual water stress conditions. However, the scatter of some points about the regression line indicates that the two methods react differently to some hydrological conditions. In contrast to TOPSIS, which measures the closeness of each alternative to the ideal solution, PROMETHEE is based on pairwise preference comparisons and outranking flows, making it more sensitive to differences between alternatives.

To further compare the rankings produced by the two methods, the rank differences were examined for the most critical wet and dry years (Table 9).

The comparison shows that the two methods do not produce identical rankings for all critical years. For these stressful years, the rank differences are relatively small, ranging from 1 to 7 positions. Among the favorable years, Among the favorable years, larger differences are observed for 2003 and 2009, with differences of 8 and 12 positions, respectively. These differences indicate that the two methods may rank some years differently despite showing a positive association in their overall results.

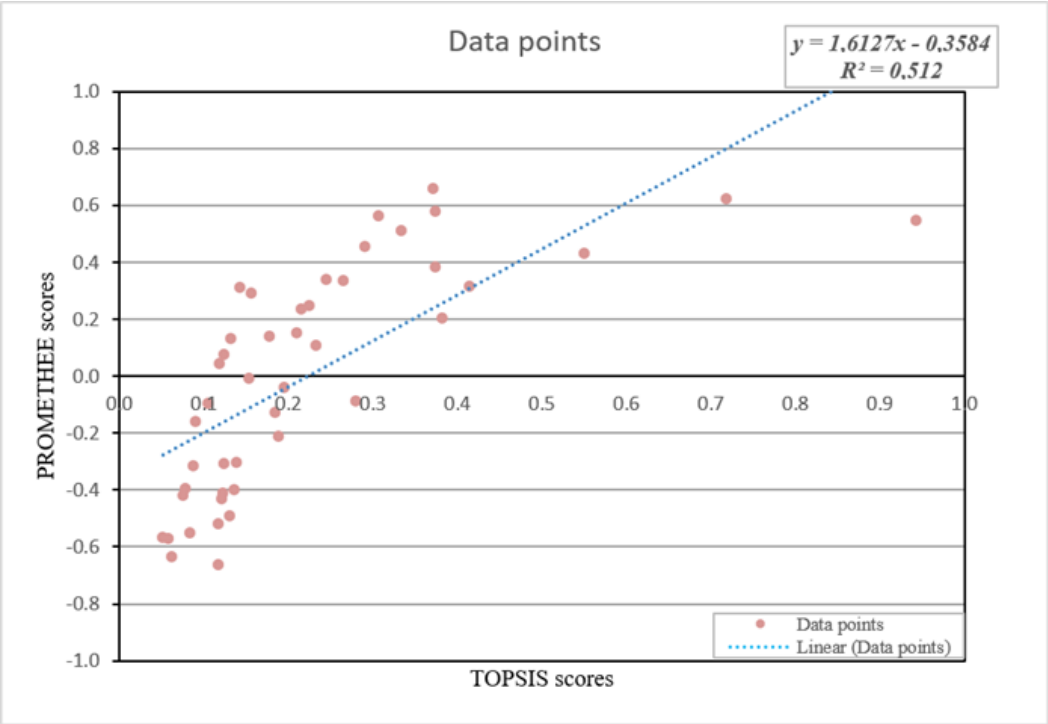


Figure 13. Relationship between TOPSIS and PROMETHEE scores.

Table 9. Comparison of TOPSIS and PROMETHEE rankings for the most critical years.

Year	Condition	TOPSIS Rank	PROMETHEE Rank	Rank Difference
2005	Stressful year	45	42	+3
2017	Stressful year	4 4	43	+1
2023	Stressful year	43	44	−1
2024	Stressful year	42	37	+5
2016	Stressful year	41	34	+7
2010	Favorable	1	5	−4
1996	Favorable	2	2	0
1997	Favorable	3	8	−5
2003	Favorable	4	12	−8
2009	Favorable	5	17	−12

Overall, the positive association between the two MCDM approaches indicates similar temporal patterns in annual water stress conditions. The observed differences in the rankings. The differences in rankings show that the two methods may rank some years differently despite their overall positive association.

5. Discussion

This study revealed that water stress in Al Hoceima Province is a structural, long-term phenomenon rather than a short-term one. Throughout the study period, water stress conditions varied substantially from year to year, as represented by the Water Stress Index (WSI), suggesting persistent pressure on available water resources. Water stress was not uniform from year to year, but there was no evidence of any long-term stabilization. However, the WSI was highly sensitive to hydro-climatic variations, with years of increased precipitation and runoff leading to temporary decreases in water stress, and dry and warm years leading to a significant increase in water stress. This indicates that climatic variability

is a key factor in water scarcity because of its direct effect on surface water availability and recharge processes. The distribution of water in semi-arid and Mediterranean systems is strongly affected by rainfall, resulting in low water availability in these zones [1]. This pattern is further accentuated by the seasonal and intermittent occurrence of rainfall [1]. The seasonal analysis reveals that water stress is strongly seasonal, with the highest WSI values occurring in summer due to high temperatures, high evaporation, and low precipitation, and lower WSI values occurring in winter because of higher precipitation and runoff generation. This means that water stress is both seasonal and structural and requires adaptive management.

The contribution analysis indicates that the most important factor affecting WSI variability is runoff, followed by precipitation. Precipitation is a fundamental component of water recharge, but its impact varies greatly depending on runoff generation and the amount of water retained [1]. Water losses are further exacerbated by increased temperature and evaporation, diminishing water availability. The influence of population on water stress increases over time but remains limited compared with the hydro-climatic factors considered in the analysis. There is general agreement between the WSI and the Falkenmark Index in identifying wet and dry periods, although differences exist because of their conceptual approaches. The WSI combines multiple hydro-climatic factors, while the Falkenmark Index looks at water availability per capita. Hence, the WSI is more responsive to short-term climatic variability, whereas the Falkenmark Index is more responsive to long-term demographic pressure. Such differences have also been observed in semi-arid and Mediterranean climatic areas with high climatic variability [46]. Thus, combining the two indices provides a better picture of water stress dynamics.

These findings are also reflected in the TOPSIS and PROMETHEE analyses. Both methods are founded on different multi-criteria decision-making principles, but the overall temporal evolution of water stress is in good agreement, and the two methods identify similar critical years. The differences among alternatives are more pronounced in the PROMETHEE scores, which show greater sensitivity to differences between alternatives, while TOPSIS scores provide a smoother representation of the annual changes. The agreement between the two methods suggests that they identify several important and favorable years in common.

Furthermore, the Innovative Trend Analysis (ITA) gives more details on the temporal behavior of the water-stress indicators. The results indicate some observations with less favorable conditions, but also relatively favorable years, in the second period of the study.

There are some relatively good years, but they are not continuous and do not change the overall long-term picture. These findings suggest that hydro-climatic variability can have a positive impact on water availability for a short period, but does not eliminate the overall increasing pressure on water resources in Al Hoceima Province.

From a management point of view, the findings suggest that seasonal variation should be taken into consideration, especially in summer when hydro-climatic conditions are more stressful and water availability is lower. Measures to enhance water storage capacity and runoff harvesting could help mitigate the effects of interannual variability. Furthermore, demand-side management remains a key component due to ongoing population growth, which puts additional pressure on water demand. Better irrigation efficiency, including the use of drip irrigation in semi-arid areas, may also help save water [47].

There are some constraints to take into account. This study does not consider direct water consumption, which restricts a full account of human water demand. In addition, reanalysis datasets may not adequately represent the local climatic features of mountainous regions due to the complexity of their topography [46]. Moreover, the WSI and the Falkenmark Index are useful integrated measures, but they reduce a complicated sys-

tem in which hydrological, climatic, and socio-economic interactions play an important role. Water scarcity cannot be captured using only one indicator [48,49]. Finally, uncertainties associated with future climate scenarios affect long-term hydrological projections in Mediterranean regions [47]. Nonetheless, using the WSI, along with the AHP, TOPSIS, PROMETHEE, and ITA methods, offers an effective framework for understanding, quantifying, and managing water stress under semi-arid conditions.

6. Conclusions

In this paper, a multi-criteria decision-making approach integrating the WSI, the Falkenmark Index, AHP, TOPSIS, and PROMETHEE was used to assess water stress dynamics in Al Hoceima Province during 1981–2025. The results suggest that water stress has been an ongoing problem in the region and is highly variable on both interannual and seasonal time scales. These findings are also consistent with recent climate projections for northern Morocco, which indicate increasing warming, drying, and persistence of climate extremes [50].

Water stress remained elevated during most of the study period, indicating persistent pressure on available water resources. The proposed MCDM framework integrates climatic, hydrological, and socio-economic components into a comprehensive assessment. TOPSIS and PROMETHEE showed similar temporal patterns, while ITA and statistical trend analysis provided additional insight into the temporal behavior of water stress. The comparison between the WSI and the Falkenmark Index also showed the value of using complementary indicators to characterize hydro-climatic variability and demographic pressure.

From a water management perspective, the proposed framework provides a decision-support tool for identifying years with severe water stress and supporting water management actions according to hydrological conditions. The findings highlight the need for adaptive water management in Al Hoceima Province, particularly during summer when water stress reaches its highest levels. Increasing water storage capacity, improving runoff collection, enhancing irrigation efficiency, and promoting efficient water use could help mitigate the impacts of climate variability. The framework could be further improved by refining the weighting system and incorporating additional information on groundwater resources, water quality, and sectoral water consumption. Climate change scenarios could also be incorporated to support long-term water resource planning in semi-arid areas.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/w18202578/s1. Table S1: WSI results under the 30 AHP weight sensitivity scenarios.

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Abbreviations

The following abbreviations are used in this manuscript:

MCDM	Multi-Criteria Decision Making
WSI	Water Stress Index
AHP	Analytic Hierarchy Process
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
PROMETHEE	Preference Ranking Organization Method for Enrichment Evaluation
ITA	Innovative Trend Analysis
CI	Consistency Index
CR	Consistency Ratio

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